

Similar Experiences, Different Outcomes: What Drives Psychotic Symptoms Isn't One Thing

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Why do some people develop psychotic symptoms while others with similar experiences do not? A new study from Karolinska Institutet suggests the answer lies not in single risk factors, but in how life experiences combine.

The research shows that substance use, genetic vulnerability, and early-life adversity do not act in isolation. Instead, their impact depends on context — meaning that the same exposure can lead to very different outcomes depending on an individual's overall life situation.

The study, conducted by Silpa Soni Nallacheruvu as part of the Master's Programme in Biostatistics and Data Science, found that risk works less like a simple on-off switch and more like a combination lock. It is not one factor, but the alignment of multiple factors, that shapes the likelihood of developing psychotic symptoms.

"Two individuals can go through similar experiences but end up in very different places," says Nallacheruvu. "What matters is how these factors come together, not just whether they are present."

The findings also reveal that different types of symptoms follow different patterns. Positive symptoms, such as hallucinations, are more sensitive to life circumstances and behavioural factors. In contrast, negative symptoms, such as social withdrawal, appear more stable and less influenced by these interactions.

This challenges a common way of thinking about mental health risk. Rather than focusing on single factors, such as substance use alone, the results suggest that risk should be understood as a broader picture — shaped by how different aspects of a person's life interact over time.

For individuals and clinicians, this means that risk is not fixed or predetermined. The same factor may carry different levels of risk depending on the surrounding context, highlighting the importance of early support and more personalized approaches to mental health.

Contact:

Silpa Soni Nallacheruvu
Master's Student, Biostatistics and Data Science
Karolinska Institutet
Email: silpa.soni.nallacheruvu@stud.ki.se

Statement on the use of generative AI:

Generative AI tools were used to assist in refining the language and structure of this press release. The content and conclusions are the author's own.